PMSCS PROGRAM

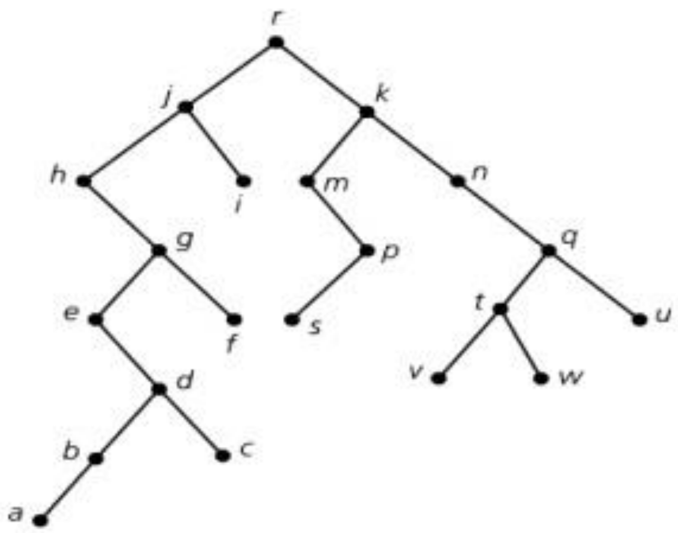
Department of Computer Science and Engineering
Jahangirnagar University

Date: 26/04/2025

Course Code: PMSCS-623P **Title:** Data Structures and Algorithms

Time: 3 Hours **TERM:** SPRING 2025 **Full Marks:** 60

Answer any SIX from the following Questions

1.	a) Analyze the running time of the following algorithm and write it using ***O***() notation: *for(int i=5; i <=(m × n); i++){* *for(int j= -155; j<=(2 × 5.5); j++){* *printf("CSE 225");* *}* *}* *for(int k=n/3; k<=(m ×n/3); k=k++){* *sum++;* *}*	**[3]**
	b) Show that $\underline{32n + 8}$ is *O(n)*.	**[3]**
	c) What do you mean by ***O(1)*** for measuring complexity of algorithm? Explain with a simple example code consisting of two loops.	**[4]**
2.	a) Briefly mention the steps of *Divide and Conquer* strategy.	**[1+2]**
	b) Explain the procedure to merge two sorted subsequences (avoiding sentinels) in case of MergeSort algorithm with example.	**[4]**
	c) Derive the complexity of MergeSort algorithm for worst case.	**[3]**
3.	a) Mention two differences between *graph* and *tree*. Explain ancestors and descendants of a *rooted tree* with example.	**[2 + 3]**
	b) Find the *post-order* traversal of the following tree.  	**[5]**

4. a) Transform the following infix expression into its equivalent postfix expression using *stack*: **[6]**

$((a \times b) / c) + (d / e) \times f$

b) Find the value of the following arithmetic expression written in *postfix* notation using *stack*: **[4]**

12, A, 1, × , /, 1, B, 5 , + , + , -

where A = alphabetic order of the first character of your first name

B = alphabetic order of the second character of your first name

Order	Letter		Order	Letter
1	a		14	n
2	b		15	o
3	c		16	p
4	d		17	q
5	e		18	r
6	f		19	s
7	g		20	t
8	h		21	u
9	i		22	v
10	j		23	w
11	k		24	x
12	l		25	y
13	m		26	z

5. a) Perform **Binary Search** for the following array named *arr = {-15, -C, -9, -4, -A, 2, B}* and *key = A*, where A = alphabetic order of the first character of your last name **[5]**

B = alphabetic order of the second character of your last name

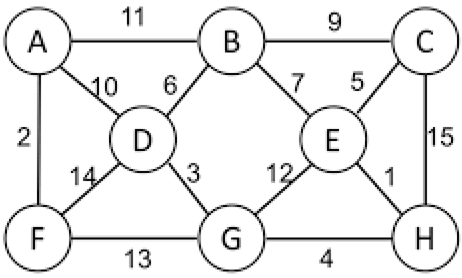
C = alphabetic order of the third character of your last name

<u>**N.B.:**</u> You can do the necessary arrangement of the given array if required before performing Binary search algorithm.

b) Perform traditional **Quicksort** algorithm to the array of Question 5. (a) to place ***B*** to its correct position. **[5]**

6. a) Apply **Buble Sort** algorithm to the array of Question 5. (a) to sort the array. **[5]**

b) Compute minimum spanning tree of the following graph using **Prim's algorithms**. **[5]**



7. a) What do you mean by prefix-free code? Explain with an example. **[3]**

b) A file has the following characters along with their frequencies: **[7]**

a:9, d:13, b:A, e:13, m:7, s:B, r: 23, q:16

where A = alphabetic order of the first character of your first name

B = alphabetic order of the second character of your first name

Draw **Huffman** coding tree and compute the *optimal* Huffman code of each character mentioned above.

8. a) Compute the optimal price for the **0-1** *knapsack* problem for knapsack capacity, **W=8**. Which items will be in the optimal solution? What will be their total price? Show how you calculate the optimal solution using a matrix containing both price and arrows for each cell in the matrix. **[6]**

item	**P**	**Q**	**R**	**S**	**T**
Weight	1	2	3	5	2
price	4	A	6	B	5

where A = alphabetic order of the first character of your first name

B = alphabetic order of the second character of your first name

b) Suppose you have to allocate a room for activities whose start and end times are as follows. Compute the largest set of activities that can be arranged in this room using a greedy algorithm. Indicate the 1st, 2nd, 3rd, … activities you choose using this algorithm and also indicate which activities you delete after you make each choice. **[4]**

i	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**
S_j	1	2	4	5	1	5	4	3	8	7	12
f_j	5	7	A	8	B	9	8	C	11	15	16

where A = alphabetic order of the first character of your last name

B = alphabetic order of the second character of your last name

C = alphabetic order of the third character of your last name

N.B. Students must submit question along with answer scripts at the end of examination.